

CRAFT: Video Diffusion for Bimanual Robot Data Generation

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craftaug.github.io

ABSTRACT

Bimanual robot learning is fundamentally bottlenecked by the cost and narrow visual diversity of real-world data. **CRAFT** (Canny-guided Robot Data Generation using Video Diffusion Transformers) synthesizes photorealistic, temporally coherent, action-consistent demonstrations by conditioning a video diffusion model on **Canny-edge control videos** extracted from simulator trajectories, a real-world reference image, and language instructions. Starting from only a handful of real-world demonstrations, CRAFT generates a large, visually diverse training set spanning **seven augmentation axes** — lighting, background, object color, object pose, camera viewpoint, wrist+third-person views, and cross-embodiment transfer — without replaying any demonstrations on real hardware (Sim2Real). CRAFT consistently outperforms augmentation-specific baselines across all conditions in simulation and real-world hardware experiments.

METHOD PIPELINE



METHOD OVERVIEW

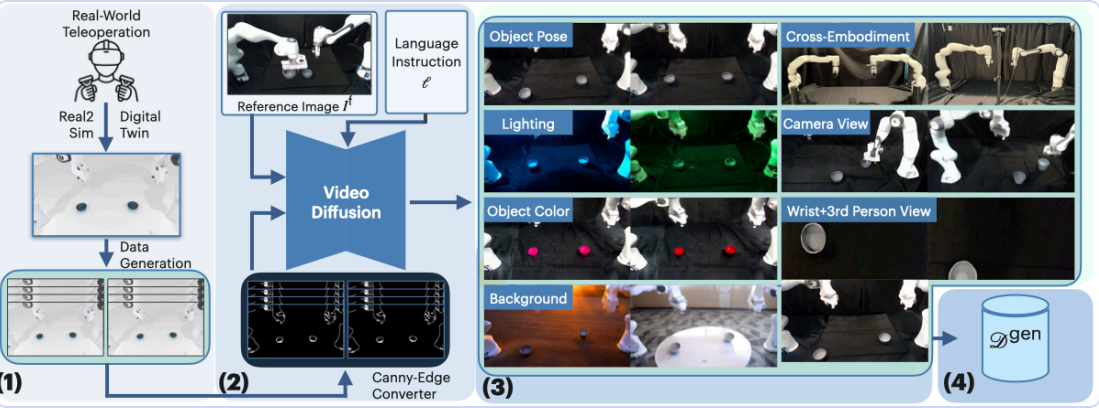
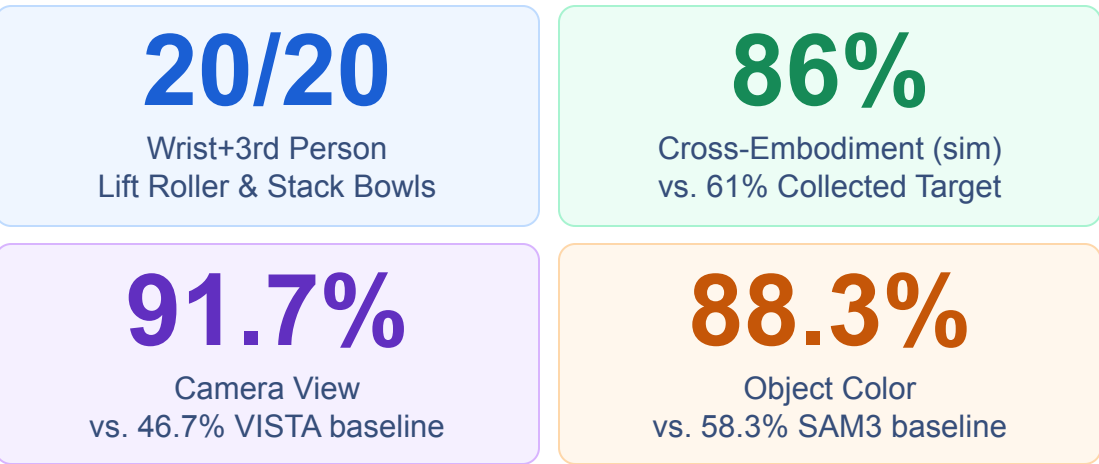


Fig. 2: (1) Real2Sim expansion $\rightarrow$ (2) Canny-conditioned generation $\rightarrow$ (3) multi-axis augmented dataset $\rightarrow$ (4) D^{gen} for ACT policy training.

WHY CANNY EDGES?

- vs. depth maps:** depth retains too much scene detail, limiting visual diversity in generation.
- vs. skeleton pose:** only captures arm structure without object information.
- Canny edges:** preserve robot-arm + object contours while discarding low-level sim texture — enabling free variation of lighting, color, and background via prompting.
- Edge thickening + connectivity post-processing bridge fragmented edges for cleaner structural guidance.

KEY RESULTS



- Contributions:** unified 7-axis bimanual data generation pipeline.
- Cross-embodiment (xArm7 $\rightarrow$ Franka) without any target-robot data collection.

SEVEN AUGMENTATION AXES

Each axis modifies either the reference image I^f , the language instruction ℓ , or the source robot trajectory — while keeping Canny control V^c fixed to preserve motion structure.

Object Pose

Decompose demo into object-centric subtasks; sample random translations and rotations over the feasible workspace in simulation; retain only successful rollouts.

Lighting

Relight reference image I^f using Veo3 under different ambient colors (blue, green, red, yellow). Shadows and reflections are preserved — unlike simple color jitter.

Object Color

Use empty-table reference image so object color is unconstrained; Canny V^c provides object contours; language prompt ℓ specifies color; LLM auto-generates a diverse color list.

Background

Omit reference image I^f entirely to prevent anchoring to the original scene; LLM-generated background descriptions in ℓ drive environment synthesis across diverse settings.

Camera Viewpoint

Place up to four cameras in simulation; tile simultaneous views into a single image; extract a tiled Canny control video preserving per-tile spatial layout in the diffusion output.

Wrist + 3rd Person

Tile left wrist, right wrist, and external camera into a 2x2 grid (fourth tile blank); mirror layout in I^f to generate synchronized, spatially consistent multi-view demonstrations.

Cross-Embodiment

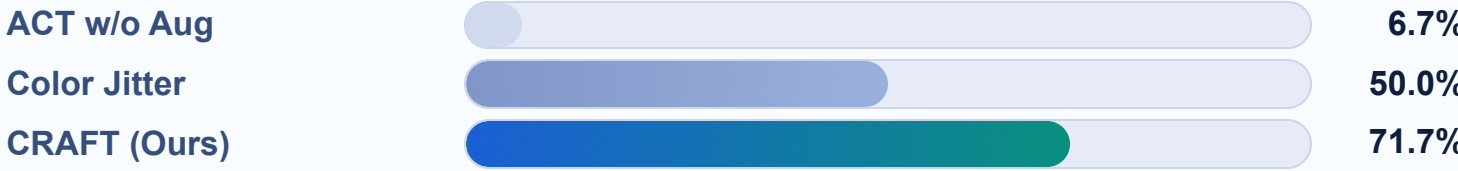
FK/IK retargets source end-effector poses to target robot joints; gripper actions preserved; retargeted trajectories replayed in sim and rendered as photorealistic target-robot videos via diffusion.

REAL-WORLD RESULTS

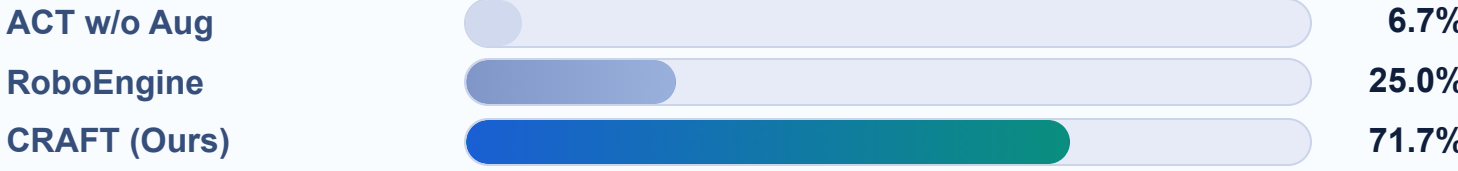
Mean success % — Lift Roller / Place Cans / Stack Bowls, 20 trials each (Table III)

● ACT w/o Aug ● Baseline ● CRAFT (Ours)

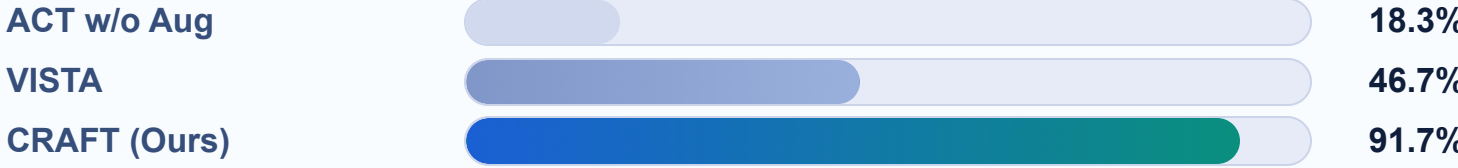
Lighting



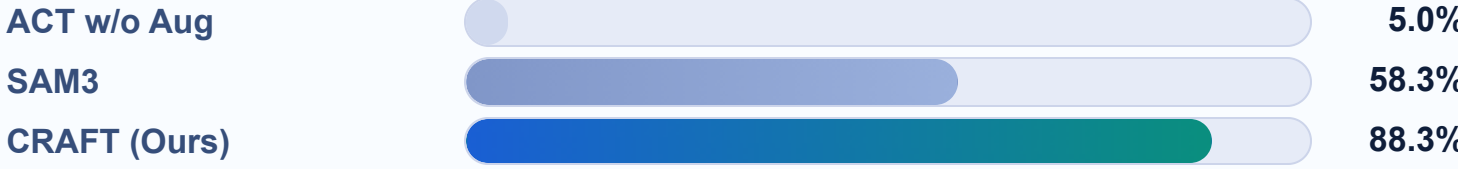
Background



Camera View



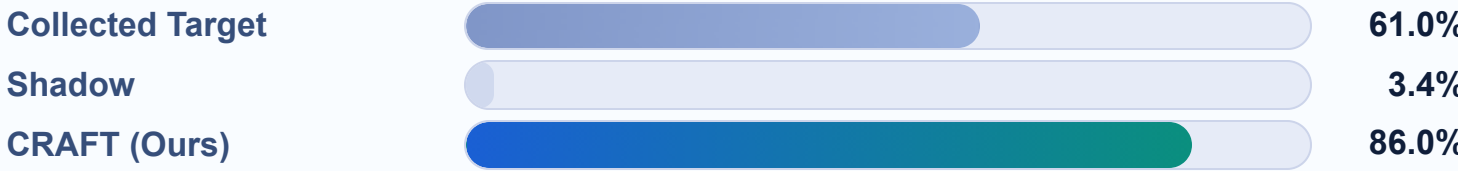
Object Color



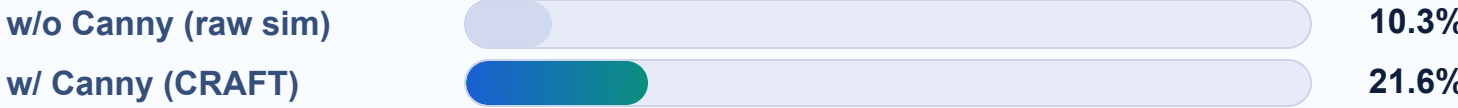
Wrist + 3rd Person



Cross-Embodiment (simulation)



Canny Ablation — Stack Bowls



Baselines: Color Jitter (Lighting), RoboEngine (Background), VISTA (Camera), SAM3 (Object Color). Canny edges nearly 2x raw sim success rate. No public baseline for Wrist+3rd Person.

CRAFT turns a small teleoperation dataset into a large, action-consistent, photorealistic training corpus — improving dual-arm robustness across seven visual axes without collecting additional real-robot data. craftaug.github.io arXiv:2604.03552v1